Fiscal Year:	FY 2007	**Task Last Updated:**	FY 01/22/2008
PI Name:	Rabin, Bernard M. Ph.D.		
Project Title:	Behavioral And Neurochemical Effects Of Exposure To Heavy Particles		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	21250-0001	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
Start Date:	06/15/2003	**End Date:**	06/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Project extended to 06/30/2007 per PI (12/06) Received NCE thru 12/31/2006, per J. Dardano (2/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Joseph, James (Human Nutrition Research Center on Aging, USDA) Shukitt-Hale, Barbara (Human Nutrition Research Center on Aging, USDA)		
Grant/Contract No.:	NAG9-1529		
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Performance Goal Text:			

Task Description:	Exposing rats to non-lethal doses of 56Fe particles produces changes in the central nervous system, particularly in dopaminergic function. Dopamine is a neurotransmitter that mediates a wide range of behaviors, including motor behavior, some types of learning, reinforcement, sensory information processing, food intake and the production of abnormal behavior patterns. Because changes in each of these behaviors can affect the performance of astronauts, the knowledge of the behaviors affected by exposure and an understanding of the neural mechanisms mediating the behaviors is essential. The proposed research program is designed to gain additional information on alterations in brain neurochemistry and behavior following exposure to particles of high energy and charge, such as those that might be encountered during space travel outside the magnetic field of the earth. The research is designed to further describe and characterize behavioral and neurochemical deficits produced by exposure to heavy particles, determine their underlying mechanisms, and develop approaches to minimize such deficits. In addition, the research program proposes to examine factors such as age and diet that may interact with the effects of exposure to heavy particles, and which may, in turn, affect the health and performance of astronauts during exploratory missions to other planets. Future missions in space (such as a mission to Mars) may involve long-term travel beyond the magnetic field of the earth, subjecting astronauts to radiation hazards posed by solar flares and galactic cosmic rays. The objectives of the research project were to describe and characterize heavy particle-induced behavioral and neurochemical deficits, determine their underlying causes, and develop approaches to minimize such deficits.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research which we have conducted previously has shown that exposing young rats (less than 3 mo. old) to low doses of heavy particles (56Fe, 600 MeV/n) has shown that the neurochemical deficits produced by this exposure are similar to those that are observed in aged rats (24 mo. old). The research which we have completed at BNL indicates that similar deficits are observed following exposure to one GeV/n 56Fe particles as well. Thus, exposing rats to low doses of 56Fe particles provides a way to accelerate aging in experimental animals with respect to certain brain and behavioral parameters so that experimenters do no have to wait for 24 months to obtain old animals. The research program is designed to understand the mechanisms by which exposure to heavy particles (primarily 56Fe) produce their effects on brain and behavior. Although the impetus for the research is to understand and minimize the effects of exposure to heavy particles on astronauts, the research program necessarily has implications for the understanding of the natural aging process. Because exposure to 56Fe particles may produce accelerated aging, this research is also indirectly concerned with the basic processes underlying the biology of aging. Similarly, because the ultimate goal of the research program is to develop interventions to minimize the effects of exposure to heavy particles on astronauts, the interventions may also prove useful with the natural aging process.
Task Progress:	Effects of Exposure to Different Particles: Based on our previous research with a range of energies of 56Fe particles, we had assumed that the acute effects of exposure to a variety of HZE particles would show the LET-dependence such that decreasing particle LET would produce deceases in the relative biological effectiveness. The results of the research reported here does not completely support that assumption. Despite the fact that the LET of 28Si (600 and 1000 MeV/n; LET ˜51 and 44 keV/µm, respectively), 48Ti particles (LET ˜106 keV/µm) and 12C (LET ˜13 keV/µm) is significantly less than that of 600 and 1000 MeV/n 56Fe particles (LET ˜189 keV/µm and 149 keV/µm, respectively), the thresholds for the disruption of neural and behavioral functioning following exposure to these particles was significantly less than that following exposure to 56Fe particles. However, this research also indicated that irradiation with different particles and energies did not produce consistent decrements in all measured endpoints. As such, the specific neural and behavioral endpoint may also be a significant factor in determining the effects of exposure to HZE particles. The data obtained thus far would suggest that it may not be possible to determine the acute effects of exposure to the range of HZE particles to which astronauts may be exposed on exploratory class missions by simply studying the effects of exposure to a single particle or measuring a single endpoint. Additional research will be needed to determine the nature of the relationship between exposure to different heavy particles and exposure-induced changes in neurobehavioral endpoints. Effects of age - longitudinal studies: Previous research had shown that exposure to a dose of 150 cGy of 56Fe particles did not disrupt the ability of the rat to respond appropriately on an ascending fixed-ratio schedule when tested several months following irradiation. However, when tested 12 months following exposure, these rats showed a disruption of operant responding. In the current series of experiments similar effects were obtained following exposure to other HZE particles, 28Si and 48Ti. Although additional research using lower doses of 48Ti particles is needed to confirm these results, the present results are consistent with the hypothesis that there is a continuing performance decrement following exposure to HZE particles as a function of the time following irradiation. Effects of age - acute studies: Previous research has shown that exposure to HZE particles produces behavioral and neurochemical changes which are characteristic of aged organisms. Because of the similar changes produced by aging and exposure to heavy particles these experiments examined the possibility that doses of 56Fe particles that were ineffective in affecting the performance of young animals would affect the behavior of older animals. The current series of experiments evaluated the effects of exposing rats of different ages (2-16 months of age) to 56Fe particles on elevated plus-maze performance (anxiety) and on amphetamine-induced taste aversion learning. In general the results indicated that lower doses of 56Fe particles were needed to affect performance in the older in animals. These results indicate that there are potential interactions between age and the neurobehavioral changes produced by exposure to heavy particles such that lower doses of these particles are needed to disrupt performance in older subjects, Since astronauts are likely to be middle aged, the results of these experiments suggest that they may be more susceptible to the effects of exposure to HZE particles on CNS function and on behaviors that depend upon the integrity of the CNS. Effects of exposure to protons: In addition to the studies of HZE particles, we have initiated studies evaluating the effects of protons on our neurocognitive endpoints. These studies were initiated because our research presented above had suggested that our failure to find significant neurocognitive effects of exposure to protons may have resulted from the use of too high doses. The results of these experiments to date are consistent with the hypothesis that previous studies on the neurocognitive effects of exposure to protons utilized too high a dose. Rats exposed to doses of 100 cGy, but not to 80 or 150 cGy of protons show a significant disruption of neurocognitive performance. These results are consistent with the results obtained following exposure to 28Si and 12C particles. Effects on recognition memory and the role of oxidative stress: The first experiment of this series was designed to

	establish the threshold dose for the disruption of recognition memory by exposure to 56Fe particles (1000 MeV/n). This experiment showed that exposing rats to 80-150 cGy of 56Fe particles disrupted recognition memory, thereby indicating that exposure to these particles produces a general disruption of different types of memory. The second experiment in this series was designed to determine the role of oxidative stress in this effect of irradiation by maintaining rats on diets containing blueberry or strawberry extract or an equivalent control diet lacking the antioxidant phytochemicals. In addition, because exposure to HZE particles occurs within a relative short time frame, in contrast to aging which occurs over an extended period of time, the rats were maintained on the antioxidant diet for only two weeks prior to irradiation. The results indicated that both 2% and 4% blueberry and strawberry diets were effective in ameliorating the effects of irradiation on recognition memory, indicating that HZE particle-induced oxidative stress mediates the effects of irradiation on neurocognitive function.
Bibliography Type:	Description: (Last Updated: 05/08/2025)
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